

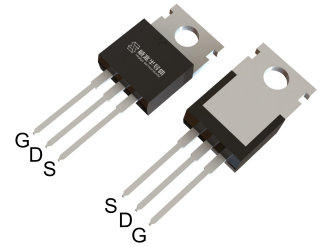
Features

- Enhancement mode
- Low RDS(on) to minimize conduction losses
- VitoMOS® II Technology
- 100% Avalanche tested, 100% Rg tested
- Optimized Qg, Qgd, and Qgd/Qgs ratio to minimize switching losses

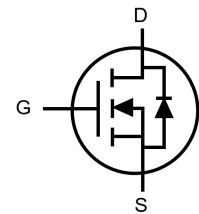


Halogen-Free

TO-220AB



Part ID	Package Type	Marking	Packing
VS11240GTH	TO-220AB	11240GTH	50pcs/Tube



Maximum ratings, at T_A =25°C, unless otherwise specified

Symbol	Parameter		Rating	Unit
V(BR)DSS	Drain-source breakdown voltage		100	V
VGS	Gate-source voltage		±20	V
IS	Diode continuous forward current (Wire bond limited)	T _C =25°C	120	A
ID	Continuous drain current @VGS=10V (Wire bond limited)	T _C = 25°C	120	A
ID	Continuous drain current @VGS=10V (Silicon limited)	T _C = 100°C	102	A
IDM	Pulse drain current tested ①	T _C =25°C	580	A
IDSM	Continuous drain current @VGS=10V	T _A =25°C	15	A
		T _A =70°C	13	A
EAS	Maximum avalanche energy, single pulsed ②		272	mJ
PD	Maximum power dissipation ③	T _C =25°C	234	W
		T _C =100°C	117	W
PDSM	Maximum power dissipation ④	T _A =25°C	2.5	W
		T _A =70°C	1.8	W
TJ,TSTG	Operating junction and storage temperature range		-55 to 175	°C

Thermal characteristics

Symbol	Parameter	Typical	Max	Unit
RθJC	Thermal resistance, junction-to-case ⑤	0.53	0.64	°C/W
RθJA	Thermal resistance, junction-to-ambient ⑥	50	60	°C/W

Electrical Characteristics

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
Static Electrical Characteristics @ T _j =25°C (unless otherwise stated)						
V(BR)DSS	Drain-source breakdown voltage	V _{GS} =0V, I _D =250μA	100	--	--	V
I _{DSS}	Zero gate voltage drain current(T _j =25°C)	V _{DS} =100V,V _{GS} =0V	--	--	1	μA
	Zero gate voltage drain current(T _j =125°C)⑦	V _{DS} =100V,V _{GS} =0V	--	--	100	μA
I _{GSS}	Gate-body leakage current	V _{GS} =±20V,V _{DS} =0V	--	--	±100	nA
V _{GS(th)}	Gate threshold voltage	V _{DS} =V _{GS} ,I _D =250μA	2	2.8	3.5	V
R _{DS(on)}	Drain-source on-state resistance ⑧	V _{GS} =10V, I _D =40A	--	4	5.5	mΩ
		T _j =100°C ⑦	--	5.9	--	mΩ
G _{FS}	Forward transconductance	V _{DS} =5V, I _D =40A	--	101	--	S
Dynamic Electrical Characteristics @ T _j = 25°C (unless otherwise stated)						
C _{iss}	Input capacitance ⑦	V _{DS} =50V,V _{GS} =0V, f=100kHz	--	3545	--	pF
C _{oss}	Output capacitance ⑦		--	1200	--	pF
C _{rss}	Reverse transfer capacitance ⑦		--	25	--	pF
R _g	Gate resistance	f=1MHz	--	1.7	--	Ω
Q _g	Total gate charge ⑦	V _{DS} =50V,I _D =40A, V _{GS} =10V	--	50	--	nC
Q _{gs}	Gate-source charge ⑦		--	16	--	nC
Q _{gd}	Gate-drain charge ⑦		--	10	--	nC
Switching Characteristics ⑦						
T _{d(on)}	Turn-on delay Time	V _{DD} =50V, I _D =40A, R _G =3Ω, V _{GS} =10V	--	15	--	ns
T _r	Turn-on rise Time		--	47	--	ns
T _{d(off)}	Turn-off delay Time		--	34	--	ns
T _f	Turn-off fall Time		--	44	--	ns
Source- Drain Diode Characteristics@ T _j = 25°C (unless otherwise stated)						
V _{SD}	Forward on voltage	I _{SD} =40A,V _{GS} =0V	--	0.86	1	V
T _{rr}	Reverse recovery time ⑦	V _{DD} =50V I _{sd} =40A, V _{GS} =0V	--	53	--	ns
Q _{rr}	Reverse recovery charge ⑦	I _{sd} =40A, V _{GS} =0V di/dt=100A/μs	--	53	--	nC

NOTE:

- ① Single pulse; pulse width $\leq 100\mu s$.
- ② This maximum value is based on starting $T_j = 25^{\circ}\text{C}$, $L = 0.5\text{mH}$, $R_G = 25\Omega$, $I_{AS} = 33A$, $V_{GS}=10V$; 100% FT tested at $L = 0.5\text{mH}$, $I_{AS} = 28A$.
- ③ The power dissipation P_d is based on $T_j(\text{max})$, using junction-to-case thermal resistance $R_{\theta JC}$.
- ④ The power dissipation P_{dsm} is based on $T_j(\text{max})$, using junction-to-ambient thermal resistance $R_{\theta JA}$.
- ⑤ Thermal resistance from junction to soldering point (on the exposed drain pad). These tests are performed on a cold plate.
- ⑥ The value of $R_{\theta JA}$ is measured with the device in a still air environment with $T_A = 25^{\circ}\text{C}$.
- ⑦ Guaranteed by design, not subject to production testing.
- ⑧ Pulse width $\leq 380\mu s$; duty cycles $\leq 2\%$.

Typical Characteristics

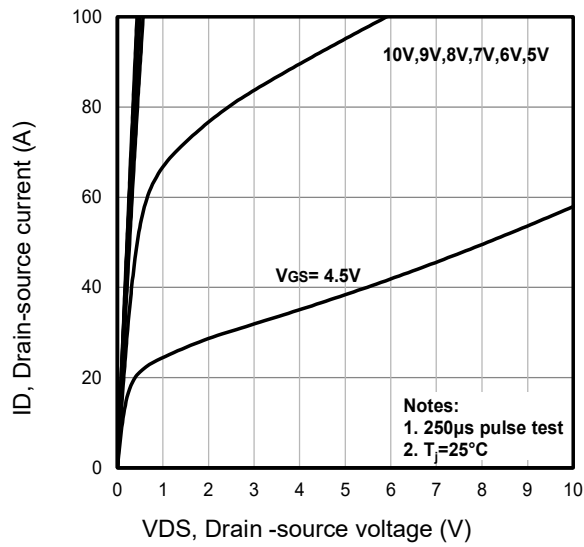


Fig1. Typical output characteristics

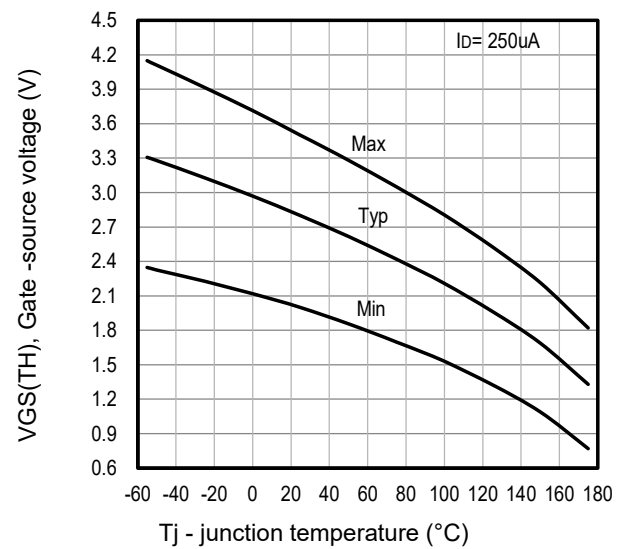


Fig2. Typical $V_{GS(TH)}$ gate-source voltage Vs. T_j

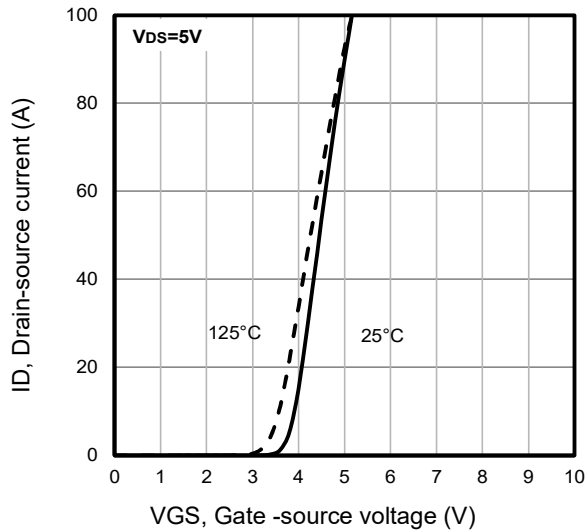


Fig3. Typical transfer characteristics

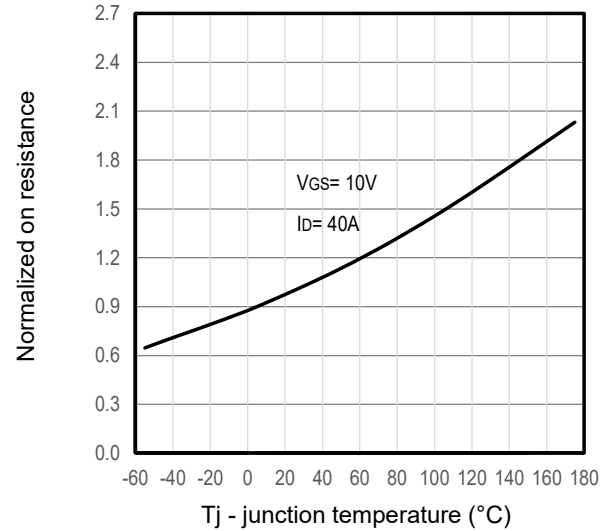


Fig4. Typical normalized on-resistance Vs. T_j

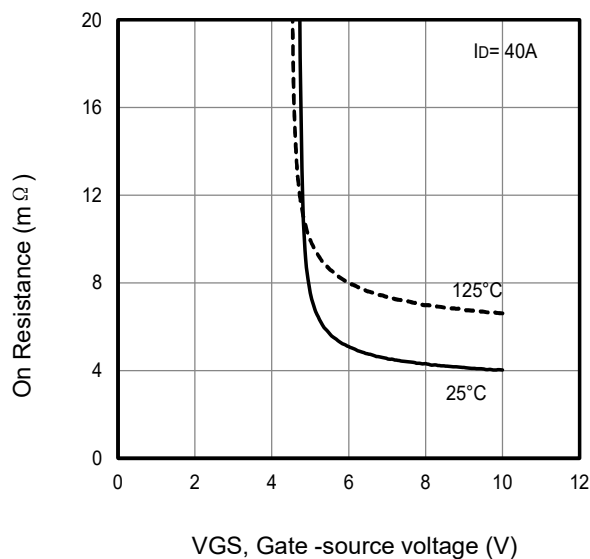


Fig5. Typical on resistance Vs gate-source voltage

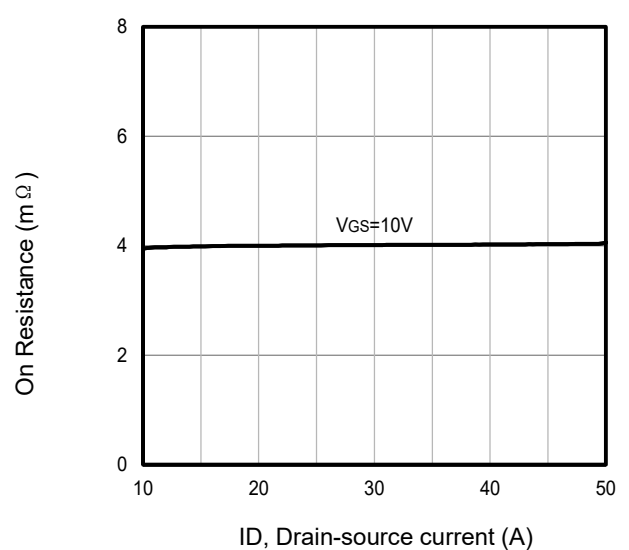


Fig6. Typical on resistance Vs drain current

Typical Characteristics

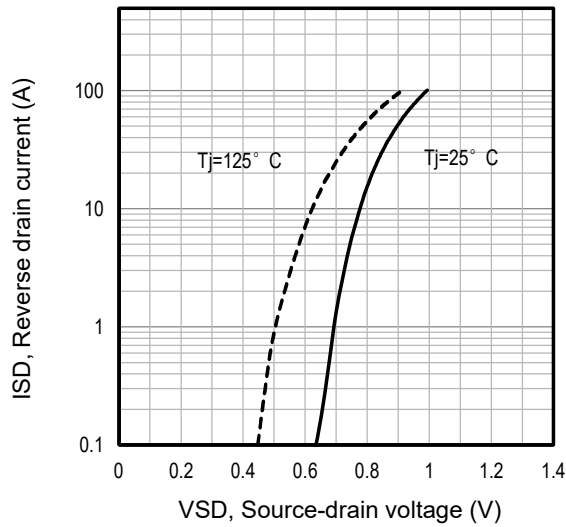


Fig7. Typical source-drain diode forward voltage

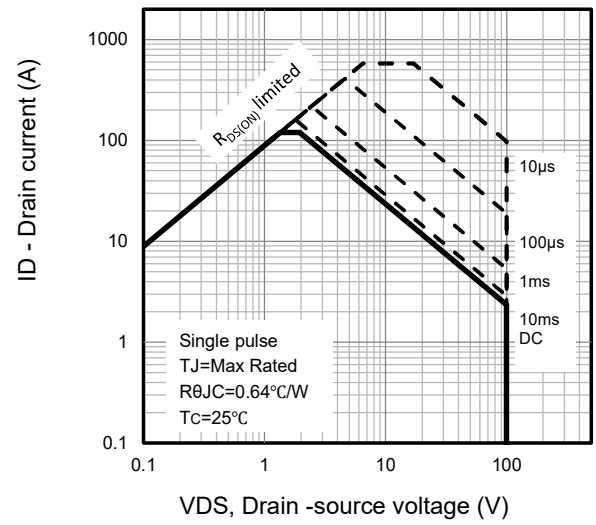


Fig8. Maximum safe operating area

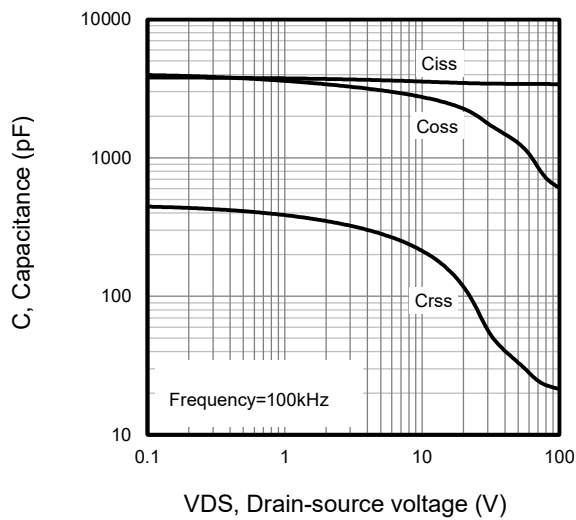


Fig9. Typical capacitance Vs. drain-source voltage

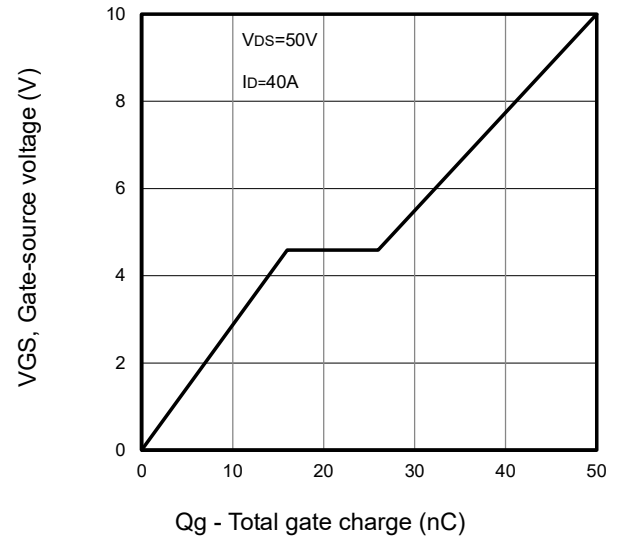


Fig10. Typical gate charge Vs. gate-source voltage

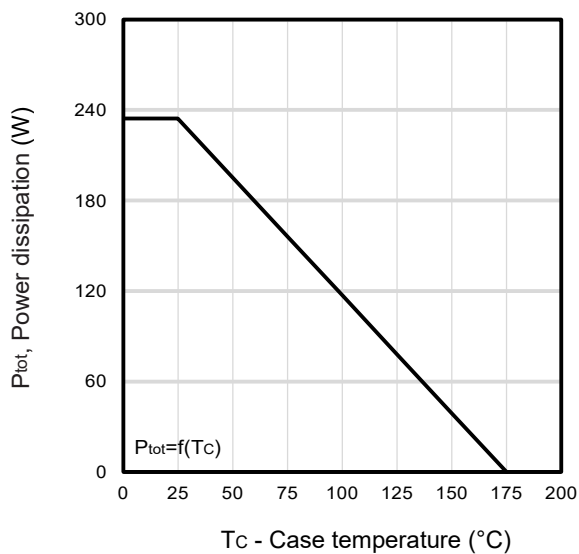


Fig11. Power dissipation Vs. case temperature

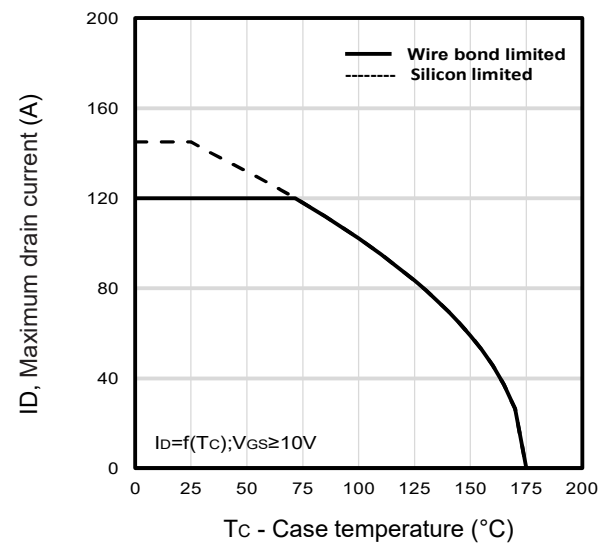
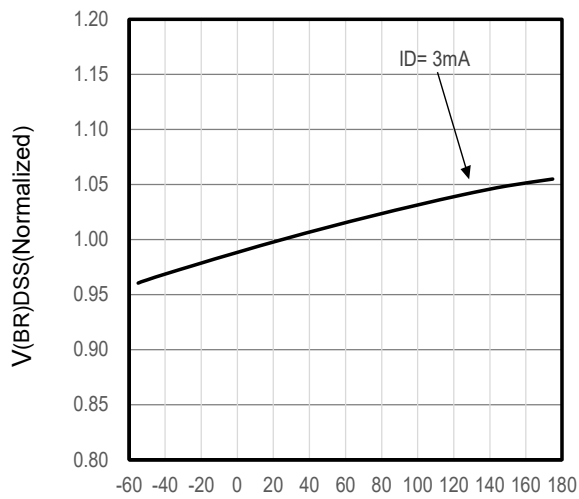


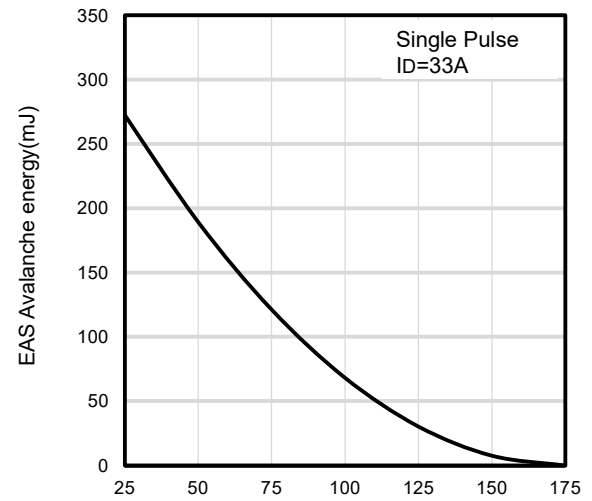
Fig12. Maximum drain current Vs. case temperature

Typical Characteristics



T_j - Junction temperature ($^{\circ}\text{C}$)

Fig13. Typical $V(\text{BR})\text{DSS}$ Vs T_j



Starting T_j junction temperature ($^{\circ}\text{C}$)

Fig14. Maximum avalanche energy vs temperature ($^{\circ}\text{C}$)

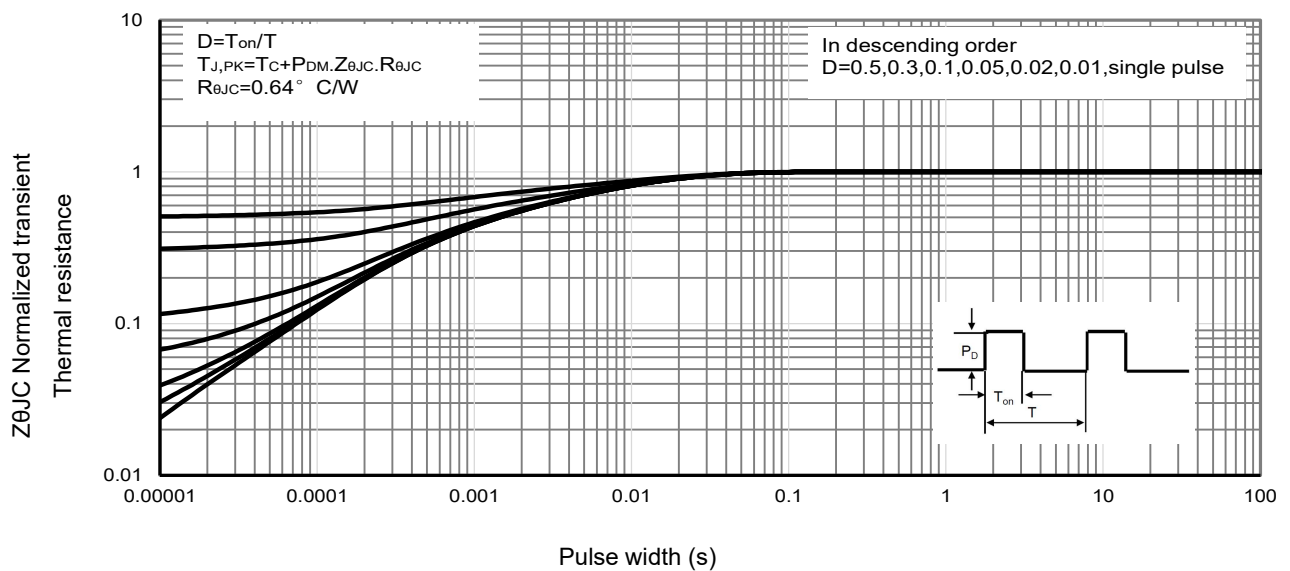


Fig15 . Normalized maximum transient thermal impedance

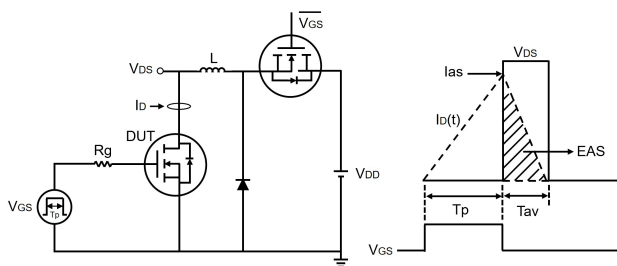


Fig16. Unclamped inductive test circuit and waveforms

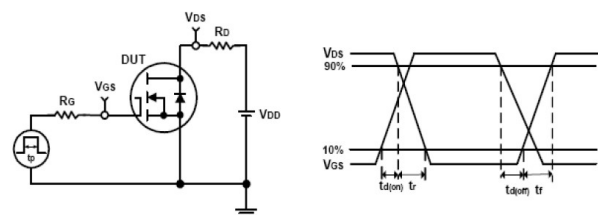
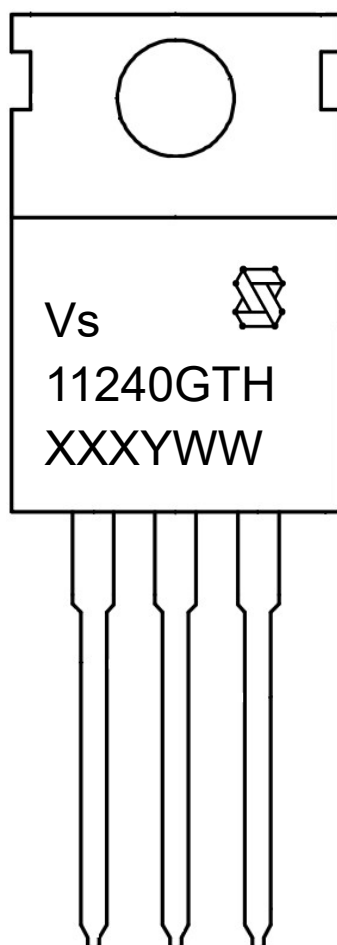


Fig17. Switching time test circuit and waveforms

Marking Information



1st line: Vergiga Code (Vs), Vergiga Logo

2nd line: Part Number (11240GTH)

3rd line: Date code (XXXYWW)

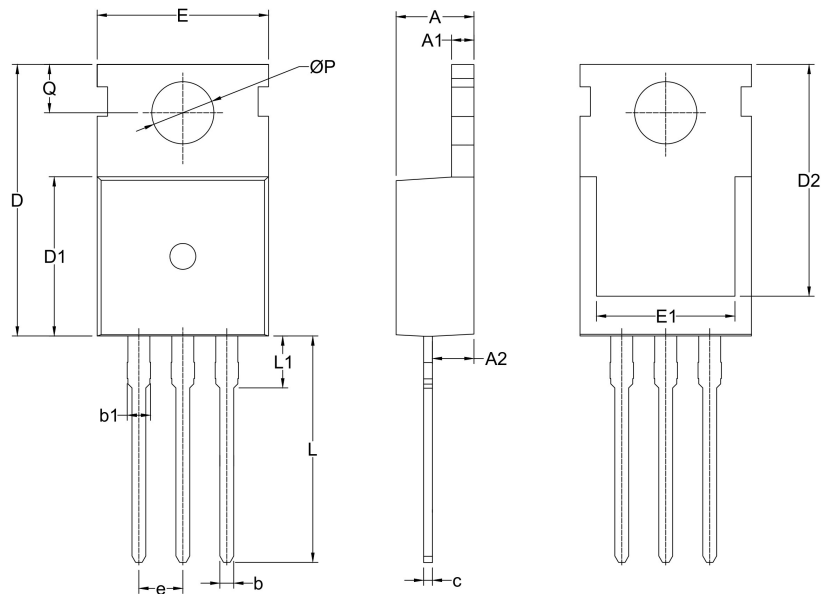
XXX: Wafer Lot Number Code , code changed with Lot Number

Y: Year Code , refer to table below

WW: Week Code (01 to 53)

Code	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T
Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030

TO-220AB Package Outline Data



Symbol	Dimensions (unit: mm)		
	Min	Typ	Max
A	4.30	4.40	4.70
A1	1.15	1.30	1.40
A2	2.20	2.45	2.70
b	0.70	0.80	0.95
b1	1.17	1.30	1.42
c	0.45	0.50	0.60
D	15.40	15.80	16.20
D1	9.00	9.20	9.40
D2	12.80	-	13.50
E	9.66	9.98	10.28
E1	7.50	-	8.00
e	2.54 BSC		
L	12.70	-	14.27
L1	2.68	-	3.40
Q	2.70	2.80	2.90
ØP	3.50	-	3.75